

Osprey 3.5 V6 low fuel rail pressure caused by high-pressure fuel pump wear

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Area: bulletins

1 Condition

Affected Osprey 3.5 V6 vehicles may store DTC P0087 (Fuel Rail/System Pressure - Too Low) accompanied by a illuminated MIL, extended cranking, and hesitation under hard acceleration. Drivers typically report a loss of power during heavy throttle when commanded rail pressure cannot be maintained. The P0087 code sets when measured high-pressure rail pressure falls below the PCM-commanded target for a calibrated period. For background on the affected delivery hardware, see the Osprey Fuel Service Manual :: Fuel Delivery .

2 Cause

The root cause is mechanical wear of the camshaft-driven high-pressure fuel pump, part number HPFP-4120, whose internal plunger and bore lose sealing capacity over high mileage. As the HPFP-4120 wears, peak rail pressure drops and the PCM sets P0087 once it can no longer compensate through pump duty cycle. Contaminated fuel and extended drain intervals accelerate this wear. Detailed pump construction and tolerances are documented in the Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) .

3 Diagnosis

Before replacing any parts, perform the structured pressure and fuel-trim workflow PROC-FUEL-DIAG to confirm that low rail pressure, and not a sensor or wiring fault, is driving P0087. Command a rail pressure test and compare actual versus desired values; a sustained shortfall under load points to HPFP-4120 wear rather than a low-pressure supply issue. Rule out sensor-side causes per the Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults . The complete diagnostic sequence is covered in the Osprey Fuel Service Manual :: Fuel Trim and Pressure Diagnosis Procedure .

4 Repair Procedure

When diagnosis confirms a worn high-pressure fuel pump, replace the HPFP-4120 following procedure PROC-FUEL-PUMP, observing the torque sequence and cam-lobe positioning to avoid damaging the new pump tappet. Always relieve system pressure and prime the rail before the first start to prevent a repeat P0087 on initial crank. Clear codes and verify commanded rail pressure is restored under a road test load. Full step-by-step replacement instructions are in the Osprey Fuel Service Manual :: Fuel Pump Replacement Procedure .